

Artificial intelligence to counteract “KPI overload” in business process monitoring: the case of anti-corruption in public organizations

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Abstract

Purpose – The nature and amount of data that public organizations have to monitor to counteract corruption lead to a phenomenon called “KPI overload”, consisting of the business analyst feeling overwhelmed by the amount of information and resulting in the absence of appropriate control. The purpose of this study is to develop a solution based on Artificial Intelligence technology to avoid data overloading and, at the same time, under-controlling in business process monitoring.

Design/methodology/approach – The authors adopted a design science research approach. The authors started by observing a specific problem in a real context (a healthcare organization); then conceptualized, designed and implemented a solution to the problem with the goal to develop knowledge that can be used to design solutions for similar problems. The proposed solution for business process monitoring integrates databases and self-service business intelligence for outlier detection and artificial intelligence for classification analysis.

Findings – The authors found the solution powerful to solve problems related to KPI overload in process monitoring. In the specific case study, the authors found that the combination of Business Intelligence and Artificial Intelligence can provide a significant contribution to the detection of fraud, corruption and/or policy misalignment in public organizations.

Originality/value – The authors provide a big-data-based solution to the problem of data overload in business process monitoring that does not sacrifice any monitored Key Performance Indicators and that also reduces the workload of the business analyst. The authors also developed and implemented this automated solution in a context where data sensitivity and privacy are critical issues.

Keywords Management control, Corruption, Artificial intelligence, Business intelligence, Big data, KPI overload, Procurement, Public organization, Internal control, Policy application, Fraud prevention

Paper type Research paper

1. Introduction

In 2018, the World Economic Forum (Johnson, 2018) and the European Parliament (The Greens, EFA Group, 2018) estimated the costs related to corruption in Public Administration, respectively, for the whole world and for the European Union, equal to 3.6 trillion and 904 billion dollars per year. It is evident that this is a relevant hit to a country's GDP. A PwC study, sponsored by the European Commission, reports that the value of public procurement in the European Union in 2010 amounted to 2.406 trillion euros: it is also shown that, analyzing the budgets of most projects, 13% of the costs incurred were attributed to corruption (PwC, 2013). Similarly, the Government Transparency Institute (Fazekas *et al.*, 2020) reports that corruption causes a loss that varies between 10 and 20% of the amount a government spends each year on procurement. The phenomenon is even more relevant in



healthcare: indeed, 7.35 trillion dollars are spent worldwide on healthcare each year and it is estimated that fraud and corruption cause a loss of 455 billion dollars. This also causes a problem of image, as 45% of citizens consider the health sector as one of the most corrupt, as reported by the Organization for Economic Co-operation and Development (NCBI, 2018). Countries that score lower in the well-known Corruption Perception Index by Transparency International are more likely to occur in scandals like these also in strategic sectors, such as healthcare: it is the case of Zambia, where 20% of the population pays bribes to access healthcare and where there was a case of misconduct in healthcare procurement that was worth 17 million dollars in losses (Nyambe, 2021). Another recent scandal regards two ex-United Nations officials who rigged some bids in Congo in the procurement of life-saving pharmaceuticals, in order to obtain millions of dollars as bribes (The Guardian, 2015). But this does not mean that the so-called “first-world” countries are exempt from cases of corruption (ECORYS and EHFNC, 2015): for instance, in 2006, a research institution in Austria was paid by a pharmaceutical company to focus its studies on a specific drug; in 2009, a relevant procurement of vaccines against the flue was conducted by the Netherlands’ government employees who were then accused of conflict of interests. More recently, corruption in Kyrgyzstan is a major factor contributing to the poor response that this country had in facing the Covid-19 pandemic, leading to problems such as mask shortage in hospitals (Satke, 2021).

To counteract corruption in the procurement process, public organizations monitor a certain set of Key Performance Indicators (KPIs). Pourshahid *et al.* (2007) recognize business process monitoring as “an emerging area which helps organizations not only plan for future improvements but also change and alter their current ongoing processes before problems happen”. Building a management framework that relies on KPIs is one of the best ways to ensure consistency with the strategic goals (Calabró *et al.*, 2015). This conclusion is reached as well by Macedo de Moraes *et al.* (2014): they study different Business Process Management life cycle models and propose a version that is consistent both with Burlton’s model and with the Association of Business Process Management Professionals’ model, which highlights the strategic and fundamental role of Process Monitoring and Controlling in the mentioned life cycle. Using KPIs tracking allows any type of organization to achieve better performance and a better understanding of the way the business is conducted but it can also be used to avoid heavy legal consequences of misconduct by identifying in advance risky phenomena and behaviors (UNDP, 2015). However, the nature and the amount of data that are needed to monitor anti-corruption KPIs make their use very difficult. The problem is not calculating these KPIs: many software tools can do that relatively easily, indeed. The problem is that these KPIs, due to their internal structure, are not punctual values: rather they are vectors or, more often, matrices built around two or more grouping variables with several levels. To control each KPI, the business data analyst needs to consider its historical trend and evolution and to decide if the values in the period under analysis are within control or not: this is what happens in business process monitoring that heavily relies on data visualization and, therefore, on human effort, such as the one based purely on Business Intelligence tools (Kang *et al.*, 2011). This method often leads to a phenomenon called “KPI overload”, which consists of having too many KPIs to control and that leads the business analyst or the organization’s managers to feel overwhelmed by the large number of performance measures that are routinely generated by the organization’s computer system and that then need to be controlled and interpreted (Stice *et al.*, 2021). This could result in one of the following scenarios:

- (1) In the short term, the organization tries to keep track of the KPIs mainly through the effort of the business analysts, thus saturating them with a workload that is hardly sustainable in the long term;
- (2) In the long term, control is lost, for some or all of the critical KPIs.

This is a specific condition of a broad issue that falls under the name of “Information Overload” or, in management and business context, “Data Overload” (Koltay, 2021), that consists, indeed, of the problem in understanding and using all the data that the organization has control of: this challenge overwhelms the analyst, as described before.

The most common solution to this problem is to reduce the amount of data, considering all the information in excess as “unnecessary” and, therefore, negligible, with the benefit of having an amount of data and information that is easier to understand and manage (Geri and Geri, 2011). This may translate into **reducing the number of KPIs to a set of extremely relevant ones**: many scholars studied how to do so, either by defining which is a reasonable number of KPIs to monitor (Osei-Owusu, 2013) or by using approaches, like the **Balanced Scorecard**, to identify the most relevant ones for a firm (Neumann *et al.*, 2012). But this of course increases the risk of **under-monitoring**, i.e. ignoring critical KPIs. In public purchasing, under-monitoring could translate into underestimating the likelihood to detect corrupt behaviors.

On the other hand, an approach based on big data analytics to monitor large amount of data could be extremely beneficial. Elkinson and McGrail (2012) discussed the potential benefits of using big data analytics and business process automated monitoring, with alerts, alarms and algorithms to detect anomalies when necessary. This implies transforming the monitoring process into what is called predictive business process monitoring, i.e. studying event logs to learn from the past and **predict new events**, for example, using deep neural network technologies (Weinzierl *et al.*, 2020). Similarly, Janiesch *et al.* (2012) study and propose a theoretical event-based framework to enhance business process monitoring. Building predictive modelling on top of process monitoring is the foundation of process mining, a completely new field with broad exploration to be conducted in order to guarantee its adoption, as described by Breuker *et al.* (2016): in particular, the authors use Petri nets to study business process behavior under the paradigm of formal languages, finding out that this technique resulted in certain biases. Furthermore, as shown by Montani and Leonardi (2010), applying predictive business process monitoring in compliance contexts (such as corruption in procurement) is particularly crucial as it allows to predict “foreseen as well as unforeseen events”, leading to an agile workflow framework. Along this direction, in the last years, some authors proposed and promoted Business Intelligence (BI) or Artificial Intelligence (AI) in business process monitoring and management control. However, only a few of them focused on the specific problem of monitoring for anti-corruption purpose: Brous *et al.* (2019) describe how statistical techniques can be used to prevent fraud cases, and Kou *et al.* (2004) identify a set of technological tools/techniques that could be helpful in fighting corruption, among which are expert systems, neural network and data mining.

However, the mentioned papers all have the same limitation. As mentioned by the respective authors, they all use algorithms that sacrifice generalizability in order to increase the precision of the predictions. In this paper, we move toward the opposite direction and wish to contribute to the literature by proposing an approach that with minimal adaptations could solve the problem of KPI overload in theoretically any context, regardless of the specific process that it is being dealt with.

Our research question, therefore, is “Can and how BI and AI be used to counteract the problem of KPI overload in business process monitoring for anti-corruption, without sacrificing any important information?”. To answer this question, we develop and engineer a new model for procurement process anti-corruption monitoring based on BI and AI, which overcomes the limitations of the current approaches. The developed solution, explicitly designed to support the Management Control department, is a tool that automatically signals situations that are out of control in a synthetic form without giving up on important information or ignoring any relevant KPI.

We developed and tested our model in a concrete scenario, the Management Control department of one of the most important Italian Scientific Hospitalization and Care Institutes (IRCCS) for organ transplantation.

2. Literature review

2.1 *The corruption problem in the procurement process and in healthcare*

Several studies agree on the fact that procurement is one of the processes that are most vulnerable to corruption and that this is even more true in the context of healthcare. [ECORYS and EHFCN \(2014\)](#) and [Vannucci \(2012\)](#) underline two different types of risk of corruption that can interfere with the procurement process:

How does

- (1) Corruption could happen in the bidding process, under the forms of collusion, bribes or, more often, requirements' design that are tailored to a specific supplier who is supposed to win the bid;
- (2) Illegal business relationship. In this case, corruption happens inside the Public Administration or, at most, between the Public Administration and the supplier and it is related to the products and services agreements in the procurement process and their quality: it could be the case of a conflict of interests.

Corruption could happen in several ways as well:

- (1) Illegally supplying public goods or services;
- (2) Conditioning public choices in order to create an advantage for the corrupter;
- (3) Giving reserved information to the corrupter in order to generate an illicit competitive advantage.

If this is corruption then lock away the whole marketing indu

The issue of corruption in Public Healthcare has been recently faced and shown in a report by the [World Health Organization \(2020\)](#). More specifically, WHO analyzes different aspects of the procurement process related to corruption:

- (1) Procurement itself. In this case, corruption implies bribes, counterfeiting goods and services agreed upon in the procurement process, preselecting the supplier that will win the bid and establishing poorly objective requirements for the bid. In particular, [collusion and bid-rigging](#) are the most common tools that corrupters use to increase prices and share reserved information;
- (2) [Pharmaceuticals purchasing](#). Risks in this case can vary from collusion, to bid-rigging, to bribery, to documents falsification, to drugs counterfeiting: the supply chain of drugs, indeed, is very susceptible to corruption, as shown by the analysis conducted in Uganda. A certain element of risk could be due to the [power of some big pharmaceutical companies](#);
- (3) Provider payments. There is, for instance, a certain risk of paying more than once for the same goods or services or paying for goods that have never been received.

Suppose now they explain

2.2 *Anti-corruption techniques and solutions*

In the very last years, a number of projects, as well as digital solutions and technologies against corruption have been launched.

A recent declaration of the Redmond-based tech colossal announces a new suite, named "Anti-Corruption Technology and Solution", whose aim is to help organizations and their stakeholders fight corruption ([Stahlkopf, 2020](#)). The suite is still in the development stage, but it was already approved by the United Nations and other relevant organizations, such as the Inter-American Development Bank. The tools used by this suite vary and include machine learning, cloud computing and other means that are useful to [detect patterns that could be proxies of corrupt behaviors](#) (see [Figure 1](#)).

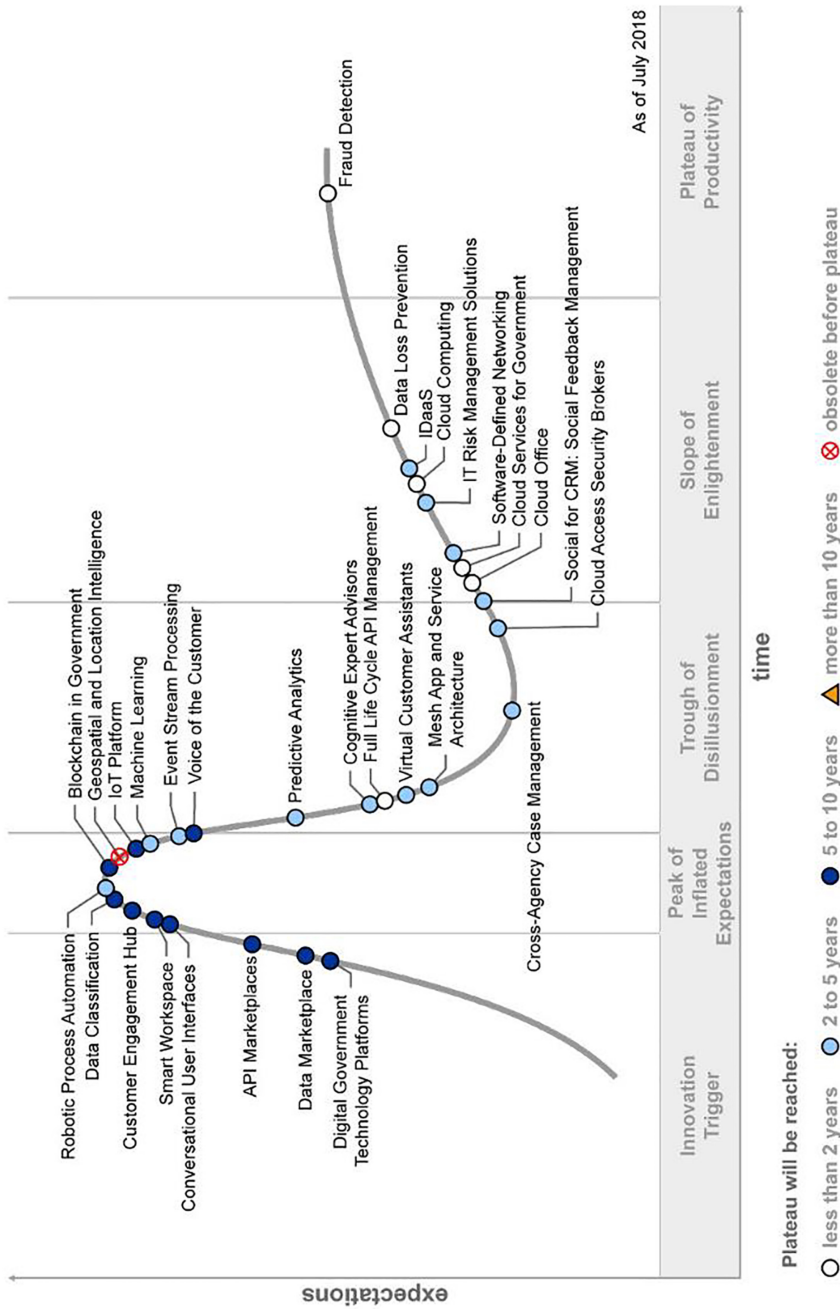


Figure 1.
Hype cycle for anti-
corruption
technologies

A recent study suggests that the return on technology investments is particularly high, especially if their aim is to use in a wise and organized way the big amount of data that organizations usually collect without using them properly; the return will be even higher in terms of reducing risks related to corruption (Santiso, 2020). The Gartner Hype Cycle, realized in 2019 for anti-corruption technologies, shows which technologies are currently used or in the development phase (Gartner, 2019).

According to Gartner, some of the most interesting technologies to fight corruption that will reach their peak in the very near future are as follows:

- (1) *Digital government technology platforms*. They are in the preliminary stage: the media are interested in these technologies, but there is no commercial release of them. Maturity will be reached between 2024 and 2030, according to the most recent estimates;
- (2) *Data classification, Blockchain, Machine learning, Internet of Things*. They are in the first development stage: other than early adopters, many other companies are trying to develop these technologies, but not everyone is succeeding. Maturity will be reached between 2024 and 2030, according to the most recent estimates;
- (3) *Predictive analysis*. It is in a stage in which many players, who tried to implement this technology, are failing and only the most innovative firms successfully bring on the idea. Maturity will be reached by 2024, according to the most recent estimates;
- (4) *Cloud computing*. It is in the stage in which this technology is gaining the trust of the general public: many investments are done in order to enhance the technology. Maturity will be reached by 2022, according to the most recent estimates;
- (5) *Fraud detection*. It is in the last stage: the technology is almost mature, successful and widely implemented. Maturity will be reached by 2022, according to the most recent estimates.

Kou *et al.* (2004) identify the following technological tools/techniques that could be helpful in fighting corruption: Finally. Maybe this should have been explained before in the article instead of saying a lot of stuff no

- (1) *Outlier detection*. The aim here is to identify some observations that are different from the standard under several parameters, therefore generating suspects of fraudulent behaviors being put in place. It's common to use this technique combined with algorithms of unsupervised machine learning, which are algorithms that do not know which patterns are related to correct behaviors and which to corrupted ones. It is also possible to use supervised machine learning, which is often more efficient but requires a bigger initial investment to create a historical database of behaviors;

- (2) *Neural networks*. They are a set of connected nodes, whose aim is imitating human brain behaviors: indeed, every node is bonded to other nodes in different layers with strong or weak bonds, based on the weight of the relationship. Therefore, every node receives a certain input and elaborates it, as long as the other nodes are linked to it: therefore, an output is generated. They can be both supervised or unsupervised; This does not explain how this is used to detect corruption!!!! What is the point to discuss the tecnolog
- (3) *Expert systems*. They are systems that support high-level decisions, being capable of managing even large amounts of data and knowledge;
- (4) *Model-based reasoning*. They are advanced statistic models;
- (5) *Data mining*. They are complex audit systems, able to analyze big amounts of data and to generate an output that the analyst can use to make decisions;

- (6) Rule-based approach. They are differential analysis systems, combined with algorithms with flexible rules and fixed control parameters;
- (7) Visualization methods. They are systems based on the human capability of recognizing anomalous patterns. Their biggest advantage is that they are based on human capabilities, intrinsically dynamic, as opposed to machine or software capabilities, usually static.

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2.3 Business intelligence and artificial intelligence in management control

One very recent work by [Arnaboldi et al. \(2021\)](#) dives deep into self-service BI tools and shows how they are useful in building informative dashboards to support management. There already are plenty of cases of good use of BI tools to monitor efficiently KPIs through some sort of alarming system: a case of success is the one of a big gas manufacturer as reported by [Alsaeedi et al. \(2021\)](#). When carefully engineered, BI tools are able to support decision-making continuously and not only at certain intervals and they are also able to be scaled and applied in many different contexts as discussed by [Rezgui and Maaouia \(2016\)](#).

On the other side, AI-based data mining approaches can definitely support the decision-making processes in management control, as shown by [Brous et al. \(2019\)](#). AI has been proven to fully express its potential through its ability to optimize existing processes and improve automation: this is, indeed, one of the first applications of AI in industrial settings which demonstrates the potentiality of these technologies for Business Process Management, as discussed by [Marrella \(2018\)](#). Other authors discuss the capability of these technologies for information transformation, but also detection, prediction and interaction with humans ([Wamba-Taguimdje et al., 2020](#)). In fact, AI and machine learning are well known in Business Process Management as well as BI: [Paschek et al. \(2017\)](#) discuss how AI and BI can be used to identify the best way to reengineer a process, [Yu et al. \(2015\)](#) show an application of Support Vector Machine to predict the values of certain KPIs in an educational setting and [Nosova et al. \(2022\)](#) discuss three types of impact that AI has on Business Process Management: virtual workforce creation, or the so-called Intelligent Automation, skills increasing and costs reduction, and innovation capacity rising.

Furthermore, [Chan et al. \(2022\)](#) show real applications of machine learning algorithms, such as Fuzzy Neural Networks, in accounting and auditing processes.

A study by [Alaka et al. \(2018\)](#) shows how these techniques can also be used in order to prevent failures, while the study by [Joshi \(2022\)](#) discusses the power that AI has in increasing the employees' productivity.

Summing up, the literature analysis demonstrates how important and relevant is nowadays to keep studying and developing solutions for counteracting the severe phenomenon of corruption; researchers in the last years have analyzed and used statistical techniques that can be used in a general fraud detection system, but also claimed the potential of AI for this purpose, given its power in automating business processes and supporting decision-making.

Our research considers these elements and aims at developing an approach based on BI and AI that solves the problem of KPI overload, without sacrificing important information while easing the effort of the decision-making process. Finally, we want to provide an example of implementation of our approach in a context that, for sensitivity and impact on society, is different from the ones usually analyzed in this field, i.e. the corruption in the procurement process of a Public Organization.

This phrase does not make sense

But until this moment the article only speaks about technologies :

3. Methodology

In this research, we aim at developing a model for procurement process anti-corruption monitoring, which overcomes the limitations of the current approaches, namely avoiding

data overload on one side and relevant data ignoring on the other side, according to the research question we stated in the introduction.

Monitoring anti-corruption in the procurement process means being able to identify in advance suspected transactions, buyers or suppliers that could be accused of fraud or corruption: the purchasing process controller has to identify these as a separate cluster from the normal transactions, buyers or suppliers. The “outlier detection” technique, as described by Kou *et al.* (2004), is based on dividing data into clusters and comparing these clusters to a “control” one, made of standard observations, considered safe: the bigger the difference between an observation cluster and the control cluster, the bigger the risk that the observation cluster is not in control. Then, to perform this type of classification analysis in the procurement process, we develop an approach based on BI, under the form of database and visualization methods, and on AI, under the form of supervised neural networks: based on the literature analysis, indeed, these tools are particularly well-suited to solve this kind of problems and, even if their application in this context is still unexplored, we consider them capable of letting us achieve all the goals we set. In this section, we describe the proposed methodology while its application in a real scenario is presented in the next section.

So the KPIs themselves

Hope they will let me know how this is done because I'm very cu

3.1 Business intelligence: automatic KPI computation and data visualization

We first build a database and use a series of SQL queries in Microsoft Access that collect data imported from an Enterprise Resource Planning (ERP) (for example, the economic value of items bought from each supplier, the economic value of items bought by each buyer, the quantities bought and so on), organize them in a proper way and use them to calculate the KPIs (for example, the economic value of items bought by one buyer from one supplier, in order to monitor if any buyer has a pattern of procurement only from a certain supplier).

Are we considering that the gathered

We will later show that in our case study the database was made of six tables with all the information related to the transactions, such as buyer data, supplier data and so on.

The calculation ends up in a so-called KPI table for each KPI. In the rows, we have the possible levels of the KPI: therefore, the number of rows of this table will depend on the number of levels of the grouping variables [1] that KPI is calculated on. The columns of this table, other than the grouping variables and the date, are the numeric values, given by the queries that are needed to calculate the KPI, first, and to control it, then. In particular, we use the Uniformly Weighted Moving Average (UWMA) control chart to perform the outlier detection analysis.

The last column [2], consequently, is a categorical variable that can take, for each row, one of the following different levels:

- (1) Situation under control, if the value of the KPI for that row is consistent with the historical trend, under the parameters of the UWMA chart;
- (2) Alarm, in the opposite case;
- (3) Uncertain/doubtful situation, if there is not a sufficient number of previous observations to infer any consideration.

As I said, historical trend may already consider corruption but ok, it's a start lets not ru

As for the rule we used to determine the control method, we want to specify that

- (1) The population is considered normal in most cases;
- (2) The fundamental parameter of the analysis (i.e. the number of periods for the moving average) is determined by balancing the number of alarms for each KPI, considering this as a proxy of the validity of the chosen value for the parameter.

This system shows several advantages:

- (1) The first is the possibility to store efficiently all the data needed for the analyses in a clean format, thus building a warehouse that is customized to the specific aim of the management control processes;
- (2) The second is that, once the setup is completed, the business process analyst does not have to calculate again the KPIs or to write again the code every month. Indeed, it will be sufficient to add the data of the new month to the database and all the results will be updated automatically, thus drastically reducing the time needed to perform the analysis. This time is even more drastically reduced if establishing a direct link between the database software that will be used (e.g. Access) and the ERP that the organization uses, thus having data that are automatically and continuously updated without any effort at all;
- (3) The third and last advantage is that the control method is customizable: either the number of periods for the moving average or the whole method of a UWMA chart as a control mechanism can easily be changed, according to the specific needs and requirements, for instance, in terms of accuracy.

Ok, so this article is really about how can we put data on a table and then draw conclusions

In our case study, we analyzed the different BI tools available in the market and chose to use Microsoft PowerBI, for which the codes have been written in DAX. But the proposed methodology can be easily applied with other data manipulation and visualization tools. Through BI software, it is possible to build and analyze databases, and through Microsoft PowerBI in particular, the SQL approach can be easily integrated. Therefore, in our case study, this tool was used only to extract synthetic information from the output of the database computations and to represent graphically the results of the monitoring.

The advantages of our BI approach are as follows:

- (1) The BI software is integrated and linked to the database: thus, updating the data in the warehouse will also update these graphs automatically;
- (2) It allows the analyst to use their human, and therefore dynamic, capabilities to analyze the data in a synthetic form, thus overcoming some of the limitations of the traditional methods as described previously;
- (3) The business process analyst is able to quickly realize which KPIs need to be prioritized and need a deeper analysis: in case of alarm, the analyst must study the transactions and determine if the alarm was justified or false. The prioritization is done simply by showing which KPIs have a greater number of alarms, thus giving the analyst every possible element to make a choice on which transactions to analyze first.

3.2 Artificial intelligence: the evolution of control

What could happen during the previous stage is that the number of alarms for some KPIs could still be particularly high and highly time-consuming to check manually (a manual check would be needed for the levels of the KPIs that are signaled as an alarm, given that the topic of corruption is particularly sensitive and needs double check): it's in this stage that the neural network algorithm can be put into place in order to reduce the monthly number of alarms for the specific KPIs that need this treatment.

The main aim of the neural network is to learn how to transform a series of statuses given by the database into a series of statuses that are consistent with what the analyst would report, filtering out the false alarms and including some alarms that were not recognized by the simple rule-based calculation of the database.

We will first describe the input that is given to the machine learning algorithm:

- (1) First, the raw input used is a spreadsheet for each KPI. In particular, this is the KPI calculation table, provided by the database queries (see [Table 1](#)). Therefore, each file presents information on the several levels of the KPI and historical data useful to monitor it, other than the categorical variable that was mentioned before;
- (2) Then, as needed, every input variable must be transformed into a numeric variable. Algorithms such as the Binary Encoding one were used in order to transform categorical text variables into a linear combination of numeric ones;
- (3) It's important to note something regarding the last input variable, i.e. the categorical variable representing the status of every level of the KPI under analysis: in this case, given the main aim of this section of the work, i.e. a filter out of false alarms, the middle level (the uncertain one) is disregarded and the other two levels are simply transformed into a binary variable.

We choose an approach based on simple machine learning, through a classification neural network. We use a supervised algorithm: this means that it is necessary to build a database of historical cases in which the output of the neural network is provided by the analyst for all the previous months. In this way, the AI algorithm will be able to learn and recognize patterns and put into practice throughout the next months what it learns, always bettering itself. The reason for which we chose to have only one hidden layer is to avoid the risks of underfitting or overfitting, which often arise in classification problems, especially given the low number of input variables that we used. Other than being widely known in the literature, this was verified empirically in the dataset we used. The number of nodes in the hidden layer was determined for each KPI according to three well-known rules given in the literature. According to these, this number must be

- (1) Smaller than the number of nodes in the input layer and bigger than the number of nodes in the output layer;
- (2) Smaller than twice the number of nodes in the input layer;
- (3) Ideally equal to 2/3 of the number of nodes in the input layer plus the number of nodes in the output layer.

We will briefly describe here other important, and more technical, elements of the neural network:

- (1) Regarding the activation function, we used a combination of ReLu and Sigmoid, in order to have an output in the form of a binary variable, exactly as wanted: the desired output, indeed, as mentioned before, is a binary variable that represents alarm (1) or situation in control (0) as the analyst would do, that meaning in a proportion that is more accurate than the one proposed by the database calculations;
- (2) As for the optimization, we used the Adam optimizer;
- (3) The general method used is the gradient descent technique;
- (4) The Binary Cross Entropy was chosen as the ideal loss function;
- (5) Finally, the last two parameters to set were the dimension of the batch and the number of epochs. In order to maximize the accuracy and to minimize the compilation time, we set the first one to 50 and the second one to 500: this choice was based on empirical results (see next section).

We will shortly describe here the main phases to program the neural network algorithm using Python code:

- (1) First of all, we use a combination of several libraries that are well known for AI programming, such as Pandas, Numpy, Matplotlib, SKLearn and Keras;
- (2) After importing the libraries and optimizing the programming environment, we prepare the dataframe, starting from the spreadsheet exported by the database, and split the columns into arrays, identifying also the input and the model for the output;
- (3) Then, the model is created and compiled, according to the parameters that we previously described;
- (4) Finally, we evaluate the accuracy of the model. We plot the accuracy and the loss function throughout the epochs, in order to evaluate the capabilities of the designed algorithm;
- (5) When we obtain satisfying results, we import a new dataframe, the one where we want the algorithm to predict the output: the first one, indeed, was necessary to train the neural network;
- (6) Lastly, we perform the algorithm on this new dataframe, thus obtaining the predictions of the status (see [Tables 2–4](#)).

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4. Case study and findings

We will now show the application of our method to a specific setting, [the healthcare institution ISMETT in Italy](#), one of the most important Italian Scientific Hospitalization and Care Institutes (IRCCS) for organ transplantation. It is a joint venture between the Sicilian Region Government and the University of Pittsburg Medical Center (UPMC) in the United States of America. We selected this organization because it is very sensitive to anti-corruption policies and innovative solutions. It is worth to mention that when we started this research project, ISMETT had already an anti-corruption system based on KPIs monitoring in place ([Pietrosi and Milli, 2014](#)). It is worth to mention that we tested our methodology only in regard to a certain set of KPIs. We did not consider all the metrics of their system, since [we considered it sufficient to test our models on a sample of KPIs](#).

Why thou

4.1 Business intelligence: automatic KPI computation and data visualization

Regarding the data we used, they are first-hand quantitative information: we used, indeed, the data collected by ISMETT in the last years (in particular, from January 2016 to August 2021) by the Management Control department, to actually put in practice what was shown up to now and to evaluate the performance.

In order to help the reader [deeply appreciate](#) the complexity of this problem, we will report here some numbers related to the organization we selected for our case study. Only referring to the year 2020 (the latest full year for which we had complete data), we count a total number of transactions equal to 9125, for a total value of 36.512.063€ corresponding to 16.018.798 articles. Each purchase order groups a number of purchase requests, so in total we have 17.959 requests. With an average monthly number of distinct buyers and suppliers, respectively, equal to 18 and 125, we obtain an average of 2.250 distinct buyer/supplier couples for each month. Regarding the database, the necessary inputs are all coming from one source, which is the ERP used by ISMETT, i.e. Oracle Fusion. Data are stored there for every transaction and purchase done by the Procurement department and every status change related to every single order is registered. This allowed us to have a broad selection of historical data available for every order, thus enabling us to easily build the necessary parts

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of the database. The output provided by the database is exactly the input needed for Microsoft PowerBI, as said before and, in our application, it is built in this form [Table 1](#):

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Table 1.
KPI table example in MS access

ID	Date	Variable A	Variable B	Numerator ^a	Denominator	KPI	Historic moving average	Historic moving standard deviation	Historic count	Status described by BI
1	15/09/2021	Supplier Example	Buyer Example	50	100	0,5	0,7	0,3	15	In control

Note(s): ^aIn our case study, the numerator would be the amount bought from the specific supplier by the specific buyer, while the denominator would be the amount bought from the specific supplier by every buyer

Source(s): Table by authors

Regarding the implementation of our model and the results we obtained:

- (1) The relational diagram for the database with all the information needed to compute the KPIs we selected is reported in [Figure 2](#).

Therefore, the database is able to associate a level of the status variable to each level of every KPI in a table form, as seen in [Table 1](#).

- (1) The data visualization tool, then, is able to make a synthesis of this table. This synthesis is represented first through a histogram for each KPI where there is a count of the observations that are categorized under the three levels of the status variables. Another step in the synthesis operation is provided by dividing the number of rows

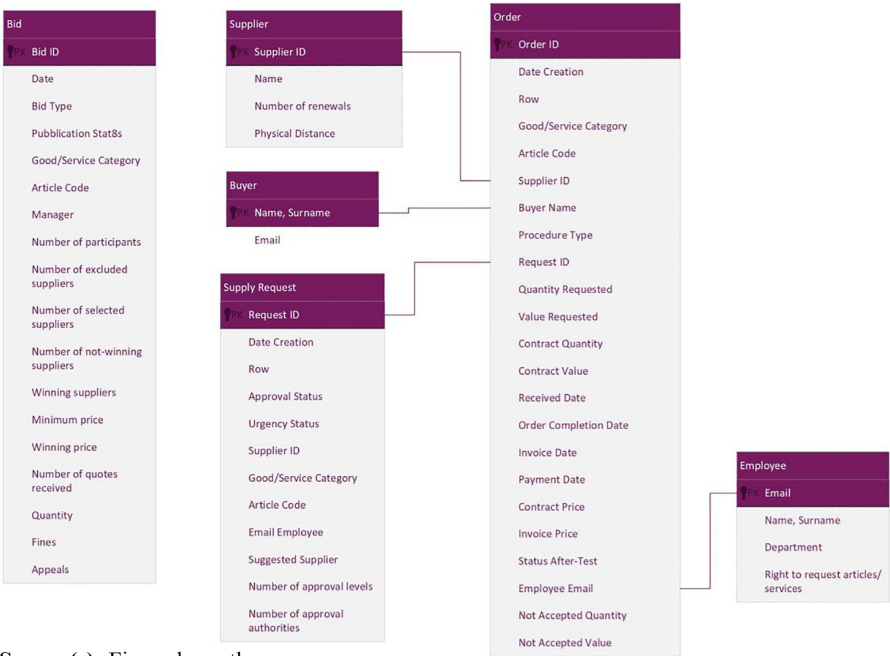


Figure 2.
Database UML
structure

Source(s): Figure by authors

under a certain status by the total number of rows of the table of the KPI: indeed, this is the percentage of alarms or uncertain situations for each KPI. It is possible to build, consequently, a histogram representing all these values in a chart: it is so possible to determine which KPIs need to be analyzed first for each category. There is a final step in synthesizing all these data: making an average of these percentages, it is possible to have an average level of alarms or uncertain situations for each category of KPIs. This way, the analyst can compare the different categories and assign them different priority levels, therefore intervening first in the most important and critical ones.

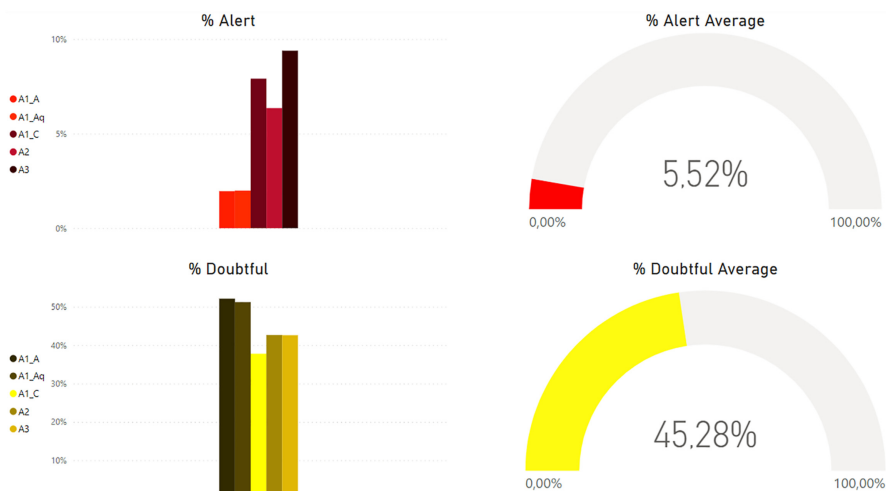
We want to present here the graphs obtained in the BI software, as mentioned:

- (1) Figure 3 shows, on the right side, the average percentages of cases of alarms and uncertain situations for a certain category of KPIs;
- (2) The left side of the same figure shows, instead, for each KPI of said category the percentages of alarms or uncertain situations;
- (3) Figure 4, finally, zooms inside a particular KPI and shows the number of rows that are categorized under each of the three levels of the status variable, together with a table with synthetic information associated with each row.

4.2 Artificial intelligence: the evolution of control

Most of the data sourcing for the AI is done by importing the output of the database, with some minimal changes, as described before. Though, the analyst needs to add a column, i.e. the status as perceived by the analyst, with real cases of alarm and of situation under control (we want to remind once again that the rows related to the “Doubtful” status are temporarily excluded from our analysis). By definition, this column should be built by an expert, in order to train the neural network with useful and real data: this would require a big investment in terms of time that unfortunately was not possible to do at the time of our study. Therefore, we created this column, filtering out some alarms where the situations seemed in control and adding alarms in situations that were originally signaled as in control by the database. This process was conducted based on a deterministic observation of the variables related to each row, with the following fundamental elements:

I hate this, we should

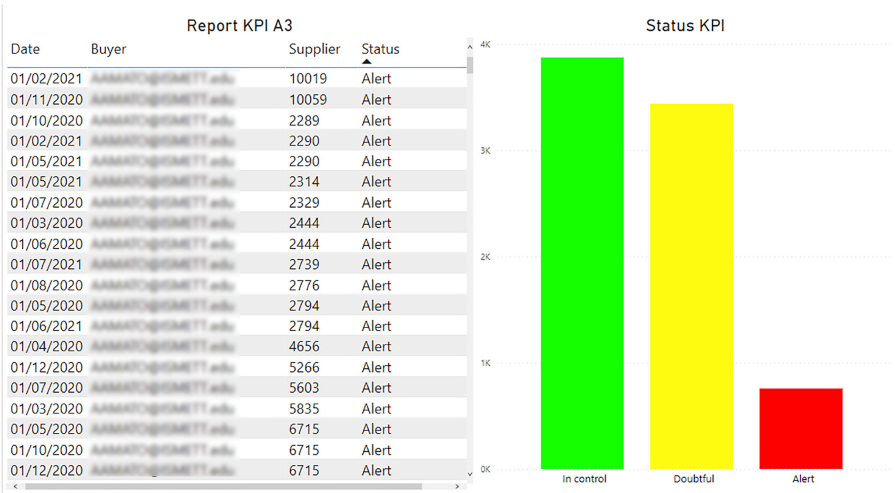


This figure is really weird a

Source(s): Figure by authors

Figure 3.
Example KPI category
synthetic status
information

Figure 4.
Example KPI status
information and detail



Note(s): The names of the buyers are covered out of privacy policies and data protection concerns
Source(s): Figure by authors

- (1) The value of the KPI;
- (2) The historical average of the KPI;
- (3) The moving average of the KPI;
- (4) The standard deviation of the KPI;
- (5) The number of past occurrences.

These rules, though, are all deterministic and are not sufficient to emulate a real (partially stochastic) scenario, therefore not justifying the implementation of an AI system: if there were no statistics involved, indeed, it would have been sufficient to refine the rules of the database in order to obtain more accurate results. Then, we used, in a small part, also some stochastic components, in order to simulate a real analysis and make the transformation more realistic and similar to what a real analyst would do.

Here is the structure of the table that will be given as an input to the neural network (see Table 2):

Date	Variable A ^a	Variable B	Numerator	Denominator	KPI	Historic moving average	Historic moving standard deviation	Historic count	Status described by BI	Status identified by analyst
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Note(s): ^aWe want to specify that, even if here they are presented as single columns, all the variables (A, B and so on), being categorical and qualitative ones, will result in a certain amount of binary columns (this amount will depend on the number of levels of each variable and on the encoding algorithm that will be used)

Source(s): Table by authors

Table 2.
Neural network input
example

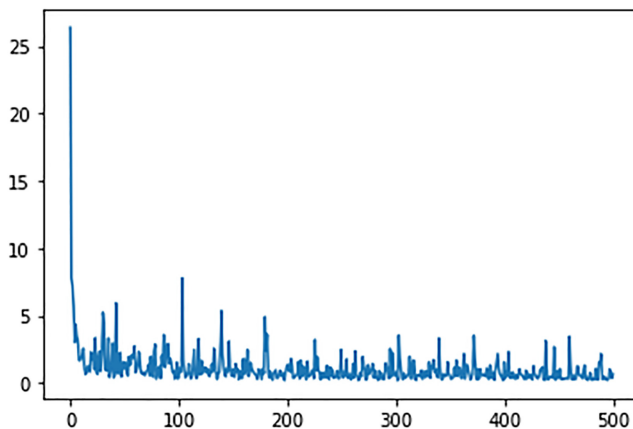
Since the neural network algorithm, coded in Python, is generic and broadly applicable, it also is the part that did not need any particular tuning related to the context, given that the main purpose of a neural network is, indeed, to be able to adapt itself to the particular setting it is used in.

4.3 Results and findings

In this section, we will describe the results and findings of our study: we will do so by analyzing the performance of our method in relation to the AI component.

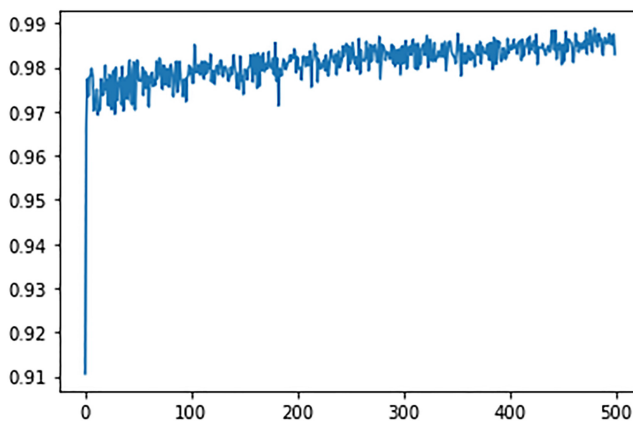
In Figure 5, we report the plot of the loss function over the epochs of the neural network algorithm, while in Figure 6, the plot of accuracy over the epochs. We present the graphs related to only one specific KPI, i.e. one of the two KPIs that aim to measure the correlation and the interdependence between buyers and suppliers, but the graphs are absolutely similar for every KPI on which we tested the neural network.

The results show that the training is completed in an accurate way: we can expect to not incur any underfitting or overfitting risk. Indeed, the loss function starts from a very high value but quickly decreases through an oscillating trend, whose width keeps on reducing over time. This means that the discrepancy between the expected outcome, as forecasted by the algorithm, and the actual outcome keeps on reducing, thus leading to an accurate estimate of the weights to assign to each node. Similarly, the accuracy keeps on increasing, reaching values that are very near to 1.



Source(s): Figure by authors

Figure 5.
Loss function plot



Source(s): Figure by authors

Figure 6.
Accuracy plot

What we just showed allowed us to evaluate the performance of the neural network that we considered satisfying. But we also wanted to understand the effectiveness of using this tool.

To do so, we built several tables, in order to evaluate the performance of our method as a whole. The first table shows each KPI:

- (1) The number of alarms detected by the simple BI;
- (2) The number of alarms detected by the BI combined with the AI;
- (3) The number of real alarms, as if they were detected by a human analyst.

Based on these values, it was possible to calculate a percentage that corresponds to the reduction of workload due to the implementation of the neural network: indeed, since each alarm needs a control in the corresponding procurement action, the higher the number of alarms is, the higher the workload results.

We present this table only for four KPIs as an example: it is possible to see that the neural network obtains results that are absolutely comparable with the reality and that the percentage reduction of the workload is almost always extremely significant. [3] (see Table 3)

This result is acc

Table 3.
Accuracy evaluation
and workload
reduction

KPI	Number of alarms [BI]	Number of alarms [BI + AI]	Number of alarms [real]	Workload reduction
A1.1	30	4	5	86,67%
A1.2	6	5	5	16,67%
A2	11	1	3	90,91%
A3	36	14	6	61,1%

Source(s): Table by authors

Finally, we build a confusion matrix. In this matrix, it is possible to identify several different conditions:

- (1) Real alarm and alarm signaled by the combined system of BI and AI. This is the most favorable case;
- (2) Real alarm but the system does not signal any alarm. This is the worst case, in which an alarm goes unseen;
- (3) Situation under control and alarm signaled by the system. This is a negative case, but not as bad as the previous one: it just leads the analyst to analyze manually a certain number of cases that will be solved as false alarms;
- (4) Situation under control and the system does not signal any alarm. This is another favorable case.

We show now an example of this matrix for one of the KPIs that we analyzed as a test. In particular, we show here the matrix for the KPI A3, given that all the other KPIs present confusion matrices that are even more accurate than this one (see Table 4):

Table 4.
Confusion matrix

	Alarm signaled	Alarm not signaled
Real alarm	5	1
Situation under control	9	313

Source(s): Table by authors

As evident, in most cases, the situation is favorable: for this KPI, 318 observations, i.e. 96.95% of the total. As a consequence, the percentage of an error incurring is particularly low, only 3.05%: we considered this risk acceptable. Furthermore, when analyzing this error, only one case out of ten corresponds to the worst case, the one of missing alarm, equivalent to 0.3% of the total observations in this case: we considered this result as extremely satisfying, given that we prefer to incur in false alarms rather than missing alarms (especially since we tested this system on KPIs that should measure the corruption risk in an organization).

To sum up, we found that the proposed solution based on BI and AI was capable of simplifying the calculation and first-screening evaluation of a set of KPIs with hundreds or thousands of levels and to do so in a relatively small time and with a high degree of accuracy. This led to a reduction of the workload of the analyst, thus minimizing the risk of “KPI overload”, with this being the main aim of our research.

5. Conclusion

In this paper, we focus on the phenomenon of corruption in the procurement process of public organizations and on the technological solutions to contrast it. On the one side, we highlighted the point of weakness of the up-to-date solutions, on the other side, we propose a method, based on a combination of BI and AI, which overcome these limitations. We implemented our method in a real case study and obtained valid and promising results.

5.1 Literature contributions

The approach shown in this paper synthesizes and builds upon the fragmented contributions of previous studies, offering an integrated solution to the problem outlined in the introduction: we are able, indeed, to gain all the previous knowledge, to enrich it and to put it actually into practice with an easy implementation but still accurate model that is ready to be used in different contexts. This paper elaborates and extends the approach proposed by [Kou et al. \(2004\)](#). We also confirm that AI can provide a significant contribution to the detection of fraud and/or corruption. In addition, we proved how they are effective also for “outlier detection” and “visualization methods”.

Furthermore, we propose a different solution to the problem of “KPI overload”, formalized by [Stice et al. \(2021\)](#) and, in general, the problem of “Information Overload”, as formalized by [Koltay \(2021\)](#). Following the approach proposed by [Elkinson and McGrail \(2012\)](#), we provide a solution that allows to control a big amount of KPIs with minimal effort and high precision, without having to give up on important metrics, especially on such an important topic, like the one related to corruption in procurement processes. Therefore, we show that reducing the amount of data, information or KPIs to track in order to be able to control a reasonable amount of data, is not the only way out: through the technological solutions and the method we propose, we show how it is possible to use big data to detect outliers, therefore enhancing the management control of an organization. This way we hope to contribute to the discussion by encouraging a new way of thinking the data, shaping all the information that an organization has control of, without having to decide what is relevant and what is negligible before even using it. Finally, the method we structure, being based both on KPIs controlling and on AI, puts into practice the theories proposed by [Calabró et al. \(2015\)](#) and [Montani and Leonardi \(2010\)](#): indeed, we build an agile holistic framework that relies on predictive business process monitoring to achieve the organization’s or department’s strategic goals. As previously mentioned, this allows the evolution of one of the five main components of the Business Process Management life

cycle, i.e. the controlling one, with evident benefits for the organization, regardless of the specific use case.

5.2 Practitioners' implications

What we found out is a concrete example of how data science tools can be powerful in every area of a business or organization. We adopted a combination of BI and AI in a specific context where these tools are not usually implemented: if it is more common to use these instruments in common management control (i.e. finance or customer-centered), it is way rarer to use them in a scenario related to corruption, where usually human work is still the force that the process of control relies on. Not only are these systems able to perform accurately and in a reliable way a kind of analysis that is very important and sensitive, but they are also able to do that automatically, thus drastically reducing the workload of those who usually perform these analyses. Indeed, our method provides businesses and public organizations with a streamlined and efficient means of monitoring a large number of KPIs, resulting in more objective and sophisticated decision-making processes: doing this with a minimal effort allows organizations to redirect human effort and time to activities that add a bigger value.

By reducing at a minimum the human work on control, BI and AI reduce also the risk of leaking sensible data, thus enhancing the respect of internal privacy policies. This is a matter of utmost importance nowadays, especially in highly data-sensitive contexts like corruption. The human intervention, at least in the final steps of the analysis, is very important because the analyst has the responsibility to decide whether or not to proceed with legal or disciplinary procedures in case of risk of corruption: so he/she needs to be very careful and to double check the few cases identified as risky by the AI before making her/his decision [4]. This concern is still ongoing and connected to the topic of ethics in using AI in decision-making processes, as largely discussed in [Lehner et al. \(2022\)](#).

Furthermore, by incorporating synthetic data visualization, the solution helps in communicating data and information between any level of the organization, regardless of the specific skill set and knowledge of the data. This last one is a feature that C-level executives must consider also in relation to communicating with external stakeholders.

Finally, the ability to effectively detect and prevent corruption or fraud has significant implications for society, both economically and in terms of broader societal impacts: we show this in relation to a specific setting, i.e. the procurement process in healthcare, but, given the nature of our approach, it is clear that what we propose can be easily generalizable to many different processes and every possible industry.

5.3 Limitations and further developments

This research has a number of limitations that we list here: at the same time, they are useful as recommendations for further developments too.

As we saw before, we used the UWMA chart, in combination with the 68–95–99.7 rule, for almost every KPI: this allows the analyst to intervene only on two control parameters. This was deemed sufficient for the purpose of this work, but other control techniques could be used in other scenarios, if judged more appropriate to the particular dataset under analysis or to the precision level that is required.

Furthermore, we used a set of pseudo-deterministic rules to create the variable of the status as if it was created by an analyst. We considered this practice sufficient for the purpose and the context of this study but in a real implementation it would be necessary to actually invest an adequate amount of time to create a historical database of suspect cases, false alarms and missing alarms: this would contribute to the training of the machine learning

algorithm, therefore obtaining results that are more consistent to the reality under examination.

Finally, as said earlier, we used the same variables exported by the database as input for the neural network. Even if this was already sufficient to obtain positive results, we suggest to use further data. In particular, adding some information for each row would help in creating a better dataset and in directing the analysis toward the direction of big data, therefore making it more powerful and accurate: we could think about demographic information, organization information, historical information and so on. In such a case, it would be possible to fully implement the power of AI, for instance, under the form of Deep Learning (instead of the Simple Machine Learning that we used).

Notes

1. The variables under which every data or process is categorized. In our case study, the grouping variables referring to each transaction will be the supplier, the buyer, the article category, etc.
2. Except for the last column [Status identified by the analyst], a sample of this table is reported in the Case Study section, see [Table 1](#)
3. We can notice how small these numbers are compared to the average number of transactions that happen in a month (presented at the beginning of this paragraph). So, the reduction we obtained through our method is even more relevant, since it brings these numbers down to a level that is manageable and easy to control for a human controller
4. This is one of the reasons why we reduce the amount of time spent by the analyst in this control process, but we do not lower it to zero.

I take from this article 2 things:

- BI can be used, instead of making t
- I hate this article but it is usefull to m
- The problem of KPI overload seems

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